



McGill

Department of
Epidemiology, Biostatistics
and Occupational Health

BIOS 640 - Introduction to Health Data Science Methods

Assignment Week 3

Instructions

1. This assignment counts for 18% of your final grade.
2. Late submissions will not be accepted
3. The totality of this assignment should be written in a single R Markdown document.
4. Your submission must include the following files:
 - The R Markdown (.Rmd) file.
 - The PDF file generated by the R Markdown file.
5. Try to adhere to the naming and styling conventions introduced during Week 1 of the course
6. Please refer to the assignment rubric to understand how your work will be graded. Good luck!

Important Notes

About the NHANES Dataset

The NHANES dataset used here is an older one, and it carries a few naming conventions that were once standard but are considered outdated today. Two are worth flagging up front:

- **gender** (originally coded as `RIAGENDR`) actually captures biological sex rather than gender identity. In modern practice, this variable would be labeled `sex`, or would allow for more than two categories.
- **ethnicity** (originally coded as `RIDRETH1` and `RIDRETH3`) describes race, a social grouping based on broadly perceived physical characteristics, rather than ethnicity, another social construct based on shared cultural and/or ancestral background. In modern practice, this variable would more accurately be called `race` or `self-reported race`.

For learning and demonstration purposes, we work with the original NHANES column labels throughout this assignment. When you clean your own datasets in practice, however, you can and should adopt clearer, more current variable naming from the outset.

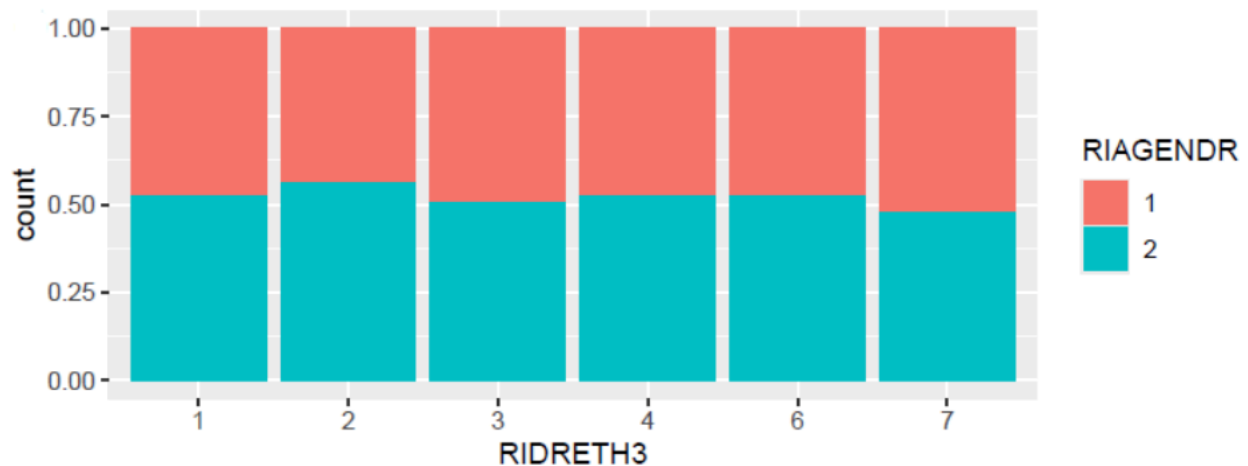
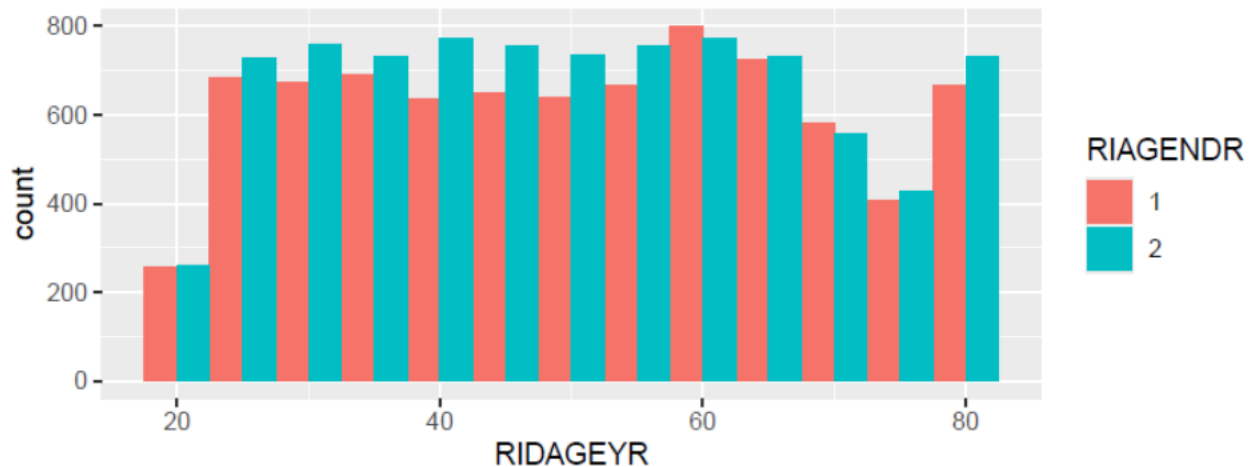
About the Figures in Exercise 1

The figures provided in Exercise 1 display the raw category codes from the original NHANES dataset (e.g., 1, 2 for gender; 1, 2, 3, 4, 6, 7 for ethnicity). Your recreated figures should display the recoded labels you assigned during assignment 2 (e.g., "Male", "Female"; the recoded ethnicity labels), since you'll be working with the cleaned version of the dataset.

Exercise 1 (20%) Reproducing and arranging ggplot2 figures

In this exercise, you will work with a cleaned version of the NHANES dataset that you used last week. It is available as `cleaned_NHANES.csv`.

- Using the `ggplot()` function, recreate the two figures displayed below. The provided figures used the original NHANES variable names and raw values. Your version of the figures should use the meaningful column names and category labels of the cleaned version. For the first figure, ensure that each bin covers 5 years.



- With your two figures, use the `cowplot` package to combine both into a single output. Then, use the `ggarrange()` from the `ggpubr` package to combine them again.
- In a short paragraph, compare `plot_grid()` and `ggarrange()`. Which layout looks better for these two plots, and why?

Exercise 2 (20%) Visualizing key characteristics

The NHANES dataset that you are currently working with has two variables representing ethnicity. In this exercise, you will explore the distribution of key variables using appropriate plots and descriptive summaries.

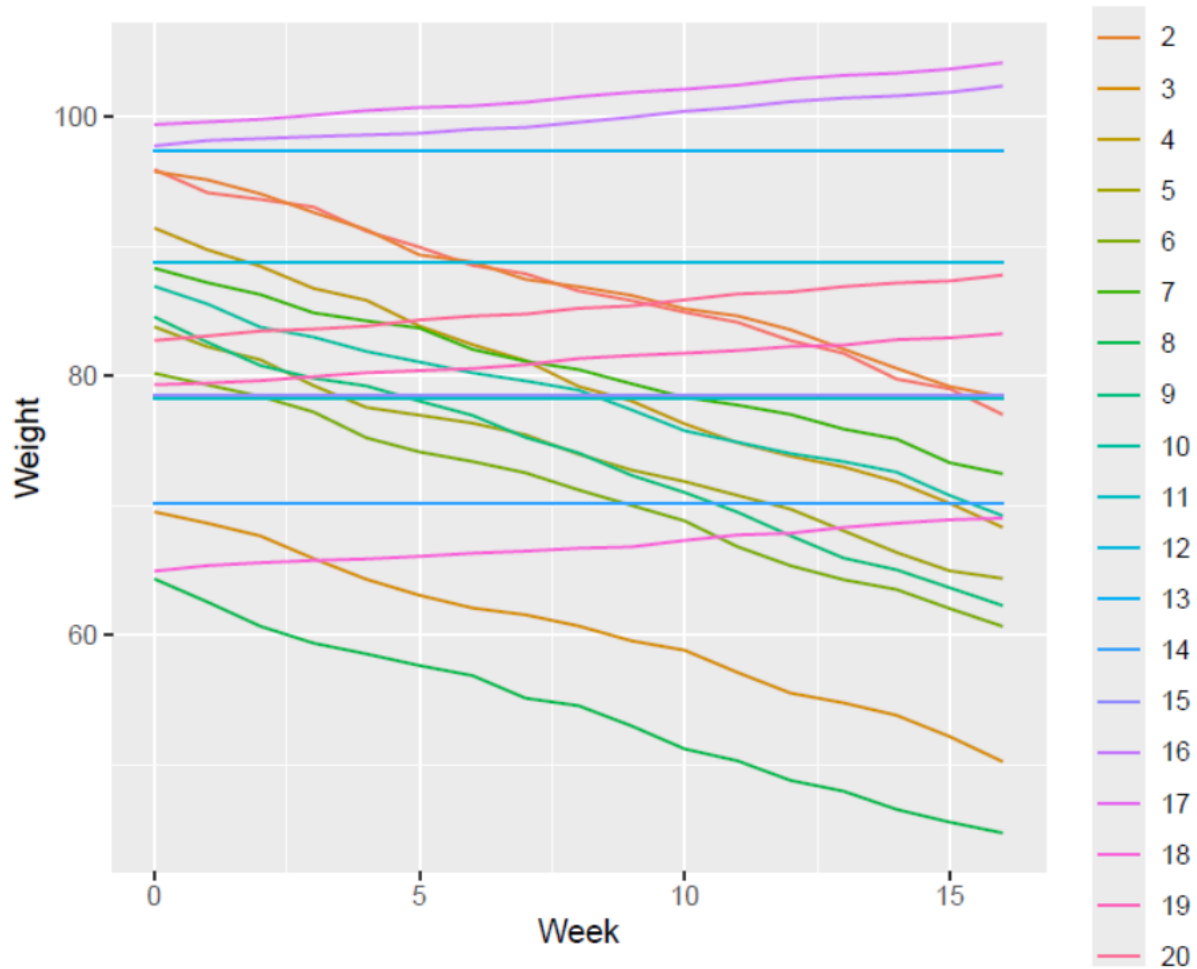
- Using `ggplot()`, create appropriate visualizations of the following variables:
 - Age
 - Gender
 - Both ethnicity variables
- Each plots should minimally include:
 - A meaningful title.
 - Axis labels.
 - A legend (if applicable).
- For each variable, briefly describe its distribution and report the number of missing values, if any. Save each of your plots using the `ggsave()` function. You will reuse one of these saved plots in Exercise 4.
- Finally, compare both ethnicity variables. Decided which is more appropriate to keep and justify your choice. Remove the variable you chose to discard.

Exercise 3 (25%) Improving ggplot figure

You have conducted a small study to evaluate the benefit of a new diet on reducing body weight over a period of 4 months. Data on participants who followed the diet are available in the file `diet.csv`.

Your goal is to create a figure that clearly shows how each participant's weight has changed over time, specifically in comparison to their baseline weight at week 0.

Your colleague rapidly generated the figure below.



- In a few paragraphs, identify and describe points that need improvement in your colleague's figure. Briefly outline how you will improve the figure and why the changes are beneficial.
- Use `ggplot2` to create a new and improved version of the figure.

Exercise 4 (35%) Exploring relationships through visualizations

The objective of this exercise is to explore the relationships between age, gender, and systolic blood pressure (SBP) using effective visual storytelling.

You will write a short report that communicates your findings. Do not refer to the numbered steps explicitly in your report. Instead, try to tell a story with a balance between text and visualization.

- Step 1: Create an average SBP variable using the 4 SBP readings. This new variable should only be missing if all four readings are missing. It will serve as your main variable of interest.
- Step 2: Use the `filter()` function to create a new datasets that includes only individuals who have non-missing values for age, gender, and average SBP.
- Step 3: Create a new categorical variable that classifies individuals based on their average SBP, using the following ranges:
 - Normal: $SBP < 120$ mm Hg
 - Elevated: $120 \leq SBP < 130$ mm Hg
 - Stage 1 hypertension: $130 \leq SBP < 140$ mm Hg
 - Stage 2 hypertension: $SBP \geq 140$ mm Hg
- Step 4: Use a maximum of four plots to describe your final sample. Make sure to describe the change in sample sizes. Consider visualizing the hypertension category counts and gender distribution. Also, reuse one saved plot from Exercise 2 and compare it to the same variable's distribution in your final sample. Briefly describe how the distribution changed (if it changed) after removing individuals with missing values.
- Step 5: Investigate how hypertension categories across age and gender using either the `facet_wrap()` or the `facet_grid()` function to display this relationship. Your final visualization must include the following elements:
 - Logically ordered categories in both panels and the legend. You can use the `factor()` function to reorder as needed.
 - A custom color palette using `scale_fill_METHOD`, where `METHOD` depends on whether you'll set the color manually or with an existing color palette.
 - A non-default theme.

Hint: Consider using the categorized version of age.